

REVIEW

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Mapping the process of digital transformation in shared services centers: a scoping literature review

Agostinho Sousa Pinto¹, António Abreu¹, Manuel Pérez Cota² and Jerónimo Paiva^{1*}

Abstract

Since the 1990s, large corporations and governments have been adopting the shared services model to reduce costs, improve the quality of services, and better manage organizational knowledge. The application of emerging digital technologies in shared services centers (SSCs) like blockchain, Cloud Computing, and Data Analytics will have a profound impact. For this scoping review, following the PRISMA protocol, 106 studies were examined to map the terrain of studies on digital transformation in SSCs. Findings point toward an undue geographical bias where most literature comes from China, and a remarkable lack of studies is observed from Southern Europe, Latin America, and Africa. Studies were categorized into seven major technological domains: Artificial Intelligence and Machine Learning (14 articles), Blockchain (9), Cloud Computing (24), Big Data and Data Analytics (25), ERP Systems (6), Robotic Process Automation (10) and Challenges in Digital Transformation and Construction of SSCs (24). This scoping review pinpoints significant research gaps in cybersecurity threats, data privacy, ethical AI regulation, and the role of Shared Services Centers (SSCs) in Sustainable Development Goals (SDGs). Future studies should address these yet-to-be-explored niches and extend their theoretical and geographical boundaries. This article serves as a starting point to advise academics and practitioners on generating safe, ethical, and sustainable digital solutions for shared services. It emphasizes that subsequent research needs to draw on broader theoretical foundations and empirical observations from underrepresented regions to facilitate equitable and secure innovations in SSCs.

Keywords Digital transformation (DT), Shared services, Emerging technologies, Big data, Cloud computing, Blockchain, Robot process automation (RPA), Enterprise resource planning (ERP), Artificial intelligence (AI), Literature review, PRISMA

Introduction

Since the 1990s, large public and private organizations have adopted the shared service model to achieve lower costs and greater efficiency [8, 34, 35, 37]. Shared Service Centers (SSCs) allow for the unbundling of administrative support functions and their centralization within a separate semi-autonomous unit. By centralizing support

activities like Finance, Human Resources (HR), and IT into a semi-autonomous unit, SSCs have become a cornerstone of modern organizational strategy for managing scale and efficiency.

The newest wave of digital transformation (DT), characterized by the creative application of emerging technologies, such as Artificial Intelligence (AI), Robotic Process Automation (RPA), Blockchain, and Cloud Computing, is fundamentally reshaping the SSC landscape [12, 13]. The very nature of SSCs—as centralized repositories for an organization's most vulnerable financial, human resources, and operations data—amplifies paramount risks in cybersecurity, data privacy, and moral

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governance [39, 40]. The integration of AI and automated decision-making systems also carries with it pressing questions of algorithmic transparency, accountability of outcomes, and the potential for inherent bias, which are egregiously unexplored in the SSC setting.

Despite the growth in the literature, there are still significant research gaps. First, the literature is highly dispersed, with research distributed across numerous technological domains without a unifying view of these interrelated risks. Second, the literature hardly addresses how SSCs can leverage digital transformation to make them drivers of sustainable value in line with the United Nations Sustainable Development Goals (SDGs). Finally, there is a pressing need to examine the exact cyber threats, ethical issues, and governance requirements required to ensure the digital transformation of SSCs is effective, secure, fair, and accountable.

Therefore, the present study makes an effort to map out the literature and to obtain a comprehensive overview of the digital transformation process in SSCs. In short, it addresses a critical research gap by examining how digital transformation in Shared Services Centers (SSCs) contributes to key UN Sustainable Development Goals (SDGs), including SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 12 (Responsible Consumption and Production). Adopting the PRISMA protocol for scoping reviews, this paper seeks to address the following research questions:

- RQ1: What is the scope and distribution of research concerning specific emerging technologies in the digitalization of Shared Services Centers?
- RQ2: What are the perceived benefits and primary operational challenges (e.g., system integration, change management, talent retention) associated with implementing digital technologies in SSCs?

- RQ3: To what extent is the literature addressing significant risks related to cybersecurity, data privacy, and ethical governance—including algorithmic transparency, accountability, and bias—in AI-enabled SSC operations?
- RQ4: How is the intersection of digital transformation in SSCs and broader corporate sustainability goals, including alignment with the United Nations Sustainable Development Goals (SDGs), addressed in the current research?
- RQ5: What are the theoretical and geographical gaps evident in the current research base, and what does this say about the generalizability of current findings and implications for future research directions?

In terms of structure, the first part will focus on exploring the concepts of “*shared services*” and “*digital transformation*”. The second part will explain the methodology based on PRISMA. Finally, the authors will summarize the main conclusions and limitations of these studies.

Digital transformation in shared services centers: a conceptual foundation

The first explicit reference to the concept of shared services was introduced by Michael J. Earl in 1994 [10]. Earl described it as a new route to efficiency. Since then, numerous definitions have emerged, but there is still no consensus among authors on a single definition. Table 1 shows a summary of the main definitions and characteristics of SSC.

Shared Services Centers (SSCs) are organizational units designed to streamline and consolidate business operations, typically focusing on back-office functions. The aim is to achieve cost savings, efficiency, and enhanced service quality through centralization [10, 49]. Table 2 provides a comprehensive overview of the main advantages associated with the Shared Services

Table 1 Key definitions and characteristics of SSCs

Definition	Key characteristics
Concentration of resources	The SSC focuses on concentrating a company's resources to process common and repetitive activities, delivering high-quality services at low cost to internal partners, pleasing customers, and adding value to the business [49]
Shared set of services	SSC was created because business units prefer sharing a set of services rather than duplicating them within the same organization [38]
Cooperative strategy	SSC is a cooperative strategy where a subgroup of existing functions is concentrated into a new semi-autonomous business unit, aiming for greater efficiency, value generation, cost reduction, and better service to internal customers while competing in the market [3]
Organizational model	SSC merges business support functions originally performed by various entities into a separate unit that provides services to the rest of the company, considering them as business customers [55]
Organizational concept	SSC consolidates processes within a company group to reduce redundancies and offers support processes. Key characteristics include: a separate organizational unit, alignment with best practices, cost-cutting as a major driver, focus on internal clients, and operation as a business [48]

Table 2 Summary of the main advantages of the SSC model

Main benefits	Sources
Cost reduction	[3, 10, 28, 38, 43, 54]
High quality	[3, 28, 43, 49]
Economies of scale	[49, 54]
Improved control	[3, 49]
Improved ROI	[3, 49]
Increased productivity	[11]
Consistency and reliability	[49, 54]
Continuous improvement	[38, 49]
Group orientation	[3, 49]
Employee motivation	[49]
Standardization	[28]
Risk minimization	[43]
Resource concentration	[49, 54]

Center (SSC) model, as identified by various authors and researchers in the field.

There is no unified definition of Digital Transformation (DT) [25]. However, DT will be defined as “a continuous process of fundamental change, driven by the innovative use of digital technologies, accompanied by the strategic utilization of key resources and capabilities, aimed at drastically improving an entity and redefining its value proposition for stakeholders” [17]. In short, the digital transformation of SSCs is a complex, continuous, and radical process that involves comprehensive changes across the technological, human, and financial aspects of a business, highlighting the need for strategic planning and execution [12, 13].

SSCs are being radically transformed by the adoption of emerging technologies—novel, fast-growing technologies characterized by a certain degree of coherence persisting over time and with the potential to exert a considerable impact [44]. Ferreira and Janssen [13] conducted a Delphi study that concluded that the six emerging technologies that would significantly impact SSCs over the next decade are: Artificial Intelligence (AI) and Machine Learning (ML); Internet/package-based automation; Business Process Management Systems (BPMS) and Robot Processing Automation (RPA); Data Analytics (DA) and Business Intelligence (BI); Blockchain and Cloud Computing [42].

According to this study, in terms of impact, AI/ML, DA and BI, and Internet/package-based automation are the emerging technologies expected to have the highest impact. Nonetheless, these technologies improve different dimensions: Blockchain, AI/ML, and Analytics are expected to improve innovation; Cloud Computing and Internet-based automation are expected to improve services and BPMS/RPA is expected to boost efficiency.

Methodology

This study employed a scoping review methodology to map the key concepts, sources, and evidence surrounding the digital transformation of Shared Services Centers (SSCs). The PRISMA protocol (Preferred Reporting Items for Systematic Reviews and Meta-analyses) was followed to carry out a systematic review and meta-analysis to ensure a transparent and reproducible process [30]. The process behind the literature review followed three discrete stages to ensure a comprehensive and systematic examination of the relevant literature [45].

Data sources and search strategy

A systematic search was performed across five major academic databases: IEEE Xplore, Web of Science, SCOPUS, OpenAlex, and JSTOR. These services were selected as they cover technology, business, and social sciences literature comprehensively. The search query employed keywords corresponding to “digital transformation” (e.g., “AI,” “blockchain,” “RPA,” “cloud computing”) coupled with “shared services center*” using Boolean operators. The search was restricted to peer-reviewed journal articles and conference proceedings in English language, published in the period 2014–2024.

Study selection and screening process

Following removal of 46 duplicates, 3,085 titles and abstracts were screened through two separate reviewers using pre-defined inclusion and exclusion criteria. A focus on digital transformation in an SSC environment through empirical, theoretical, or case studies was required to be included. Exclusion criteria removed non-peer-reviewed manuscripts, theses, dissertations, irrelevant publications, and unavailable full texts. Disputes were resolved through discussion or third-party arbitration. One hundred sixteen articles were retrieved for full-text review in this stage. Following rigorous screening, 106 articles met all the inclusion criteria, with exclusions primarily due to insufficient focus on SSCs or failure to complete digitization details.

Figure 1 shows the PRISMA Flowchart that summarizes the entire study selection and screening process.

Data analysis

The data analysis of the 106 included articles was both quantitative and qualitative:

- **Bibliometric Analysis:** VOSviewer, a bibliometric software, was utilized to conduct a co-occurrence keyword analysis to identify key research themes and conceptual clusters within the literature.

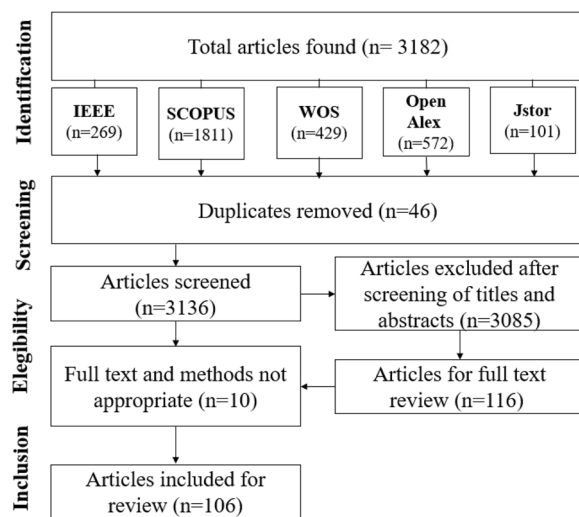


Fig. 1 PRISMA flowchart

- **Thematic Synthesis:** After conducting bibliometric mapping, a systematic thematic analysis was performed. This process facilitated the identification of prominent research topics, gaps, and connections between various technological domains and their implications.
- **Descriptive Statistics:** Microsoft Excel was also used to create helpful charts and tables using the descriptive data (e.g., publication year, geographical focus, research methods) and extracted and tabulated to provide an overview of the research landscape.

Main results

This section presents the bibliometric and thematic results of the literature review of SSC digital transformation. Results are structured to progress from general research topics to specific insights related to development and theoretical foundations.

Analysis of the main keywords

In Fig. 2, it is possible to create a co-occurrence analysis of the main keywords created with VOSviewer. The node area and font size are proportional to the number of occurrences of a certain keyword, thus larger nodes reflect the keywords that were mentioned more often. The line between nodes indicates when a keyword appears with another keyword. Of the total of 289 keywords detected, only 54 meet the minimum threshold of 2.

Table 3 provides an overview of each cluster encompassing several relevant aspects of digital transformation. The analysis identified a total of seven keyword clusters within the field of digital and technological advancements. Each cluster is identified by a distinct color and focuses on a specific area of study.

Analysis of publication growth

In Fig. 3, it is possible to see a graph displaying the number of articles between the years 2014 and 2024. Results have shown that the number of articles has grown significantly.

Geographic distribution of research

With the adoption of the shared service model by Chinese conglomerates (Tencent, China General Nuclear

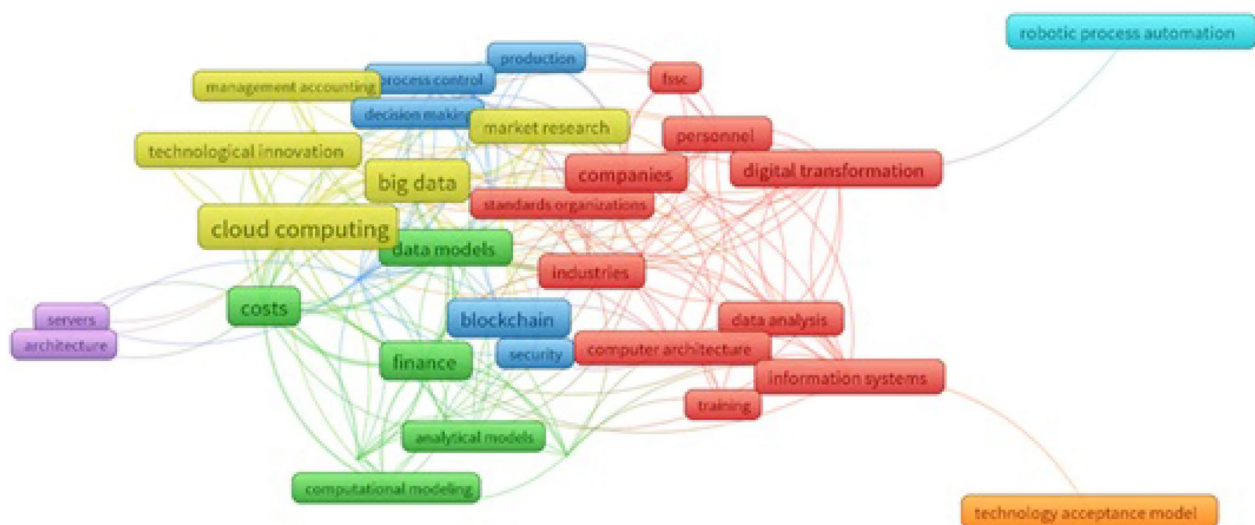


Fig. 2 Co-occurrence map of keywords made in VOSviewer

Table 3 Cluster topics and keywords

Cluster	Topic	Main keywords
Red	Digital transformation	Digital transformation; companies; industries; personnel; fssc; data analysis; information systems; training
Green	Financial and cost analysis	Costs; Finance; data models; analytical models; computers; predictive models
Blue	Blockchain and security	Blockchain; security; decision making; financial sharing; production
Yellow	Big Data and cloud computer	Big data; cloud computing; technological innovation
Purple	IT architecture and infrastructure	Architecture; authentication; servers
Light blue	Automation technology	Robotic process automation; automation
Orange	Technology acceptance model	Technology acceptance model; perceived ease of use

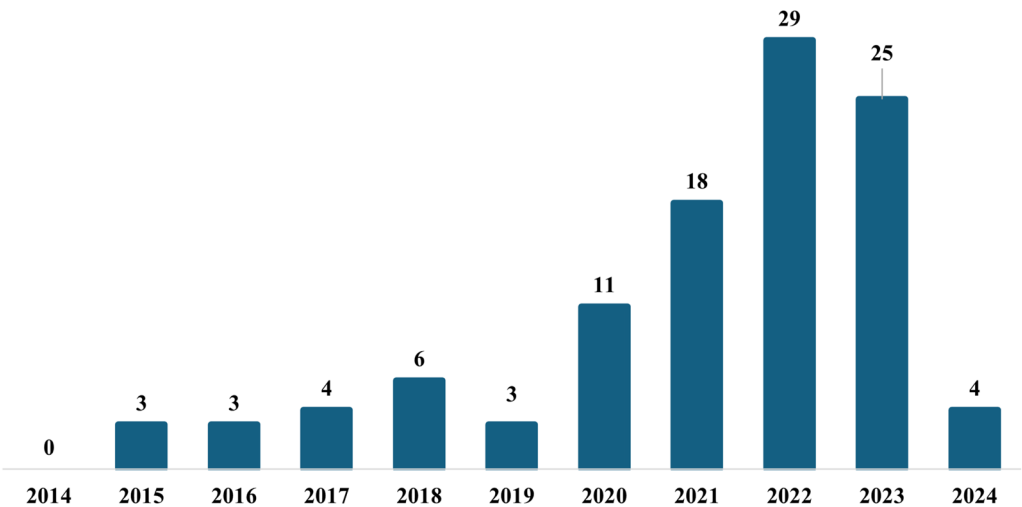


Fig. 3 Number of articles published between 2014 and 2024

Power Cooperation, and ZTE), the emergence of global business units, and mastery of emerging technologies, China is at the forefront of advancing knowledge on the topic of digital transformation of SSCs (70 articles). Grachová explained that the Chinese government sought to incentivize the creation of SSCs in tier 2 and 3 cities (Dalian, Tianjin, Wuhan, Chengdu, Xi'an) because tier 1 cities (Beijing, Shanghai, Guangzhou, and Shenzhen) have become overcrowded [18].

Figure 4 represents the geographical distribution of the research.

Due to a combination of strategic, technological, and business factors, Eastern Europe has emerged as one of the most dynamic regions for SSCs. As a result, it is possible to explain the strong presence of articles from Eastern European nations, especially Poland (9 articles). According to ABSL, Poland's mixture of highly educated and multilingual human capital, significant cost advantages, and a developed, innovation-oriented environment solidifies its status as a premier destination for SSCs [41]. In 2023, the business services sector

accounted for approximately 5.3% of Poland's GDP, with 457,100 employees in the sector and 1941 centers operating across Poland.

Key research institutions and authors

Table 4 shows the most productive research institutions in terms of publishing scientific articles, most of which are from China, ordered in descending order by the number of articles published.

Focusing on the topic of authorship, out of 208 authors, only four wrote more than one article and none of them co-authored two or more articles. In terms of affiliations, it is worth noting that no institution produced more than 4 articles. The distribution of 106 articles across 77 different journals and 29 different conferences, with no single venue publishing more than three articles, reveals a highly dispersed body of knowledge. As a result, it is possible to conclude that research into this topic is interdisciplinary and of broad interest to researchers in fields like Management, Information Systems, Engineering, and Computer Science.

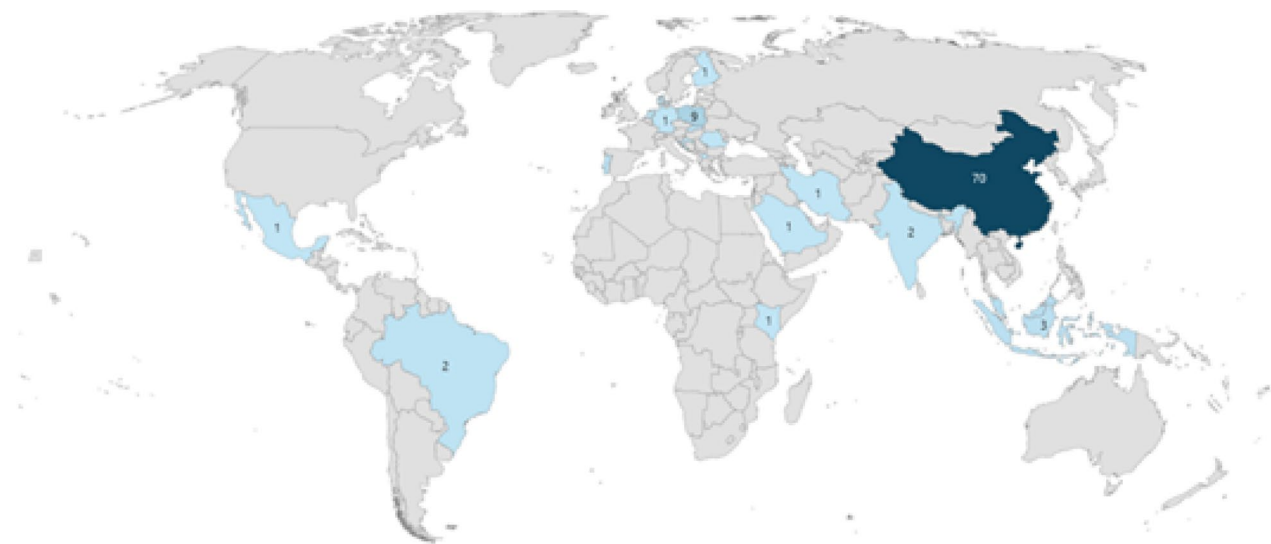


Fig. 4 The number of published articles by country

Table 4 Most productive research institutions

Institutions	Number of articles
The Seventh Metallurgical Construction Corp. Ltd	4
University of Bratislava	3
Beijing Institute of Technology	
Nanjing University of Science and Technology	
Sichuan International Studies University	
Shanghai University	
Nanjing University	
Yunnan University	

Table 5 Theoretical frameworks used

Framework	References	n
Resource-based View (RBV)	[22, 29, 57]	3
Dynamic capabilities (DCT)	[22, 29]	2
Transaction cost economics (TCE)	[29]	1
Theory of planned behavior (TPB)	[31]	1
Organizational support (POS)	[51]	1
UTAUT	[31, 51]	2

Theoretical frameworks

Table 5 shows the theoretical frameworks used in the publications. It is worth noting that only three publications were based on Resource-based View (RBV), two publications were based on Dynamic Capabilities Theory (DCT) and Unified Theory of Acceptance and Use

of Technology (UTAUT), one publication was based on Transaction Cost Economics (TCE), Theory of Planned Behavior (TPB) and Perceived Organizational Support (POS).

Main research themes

To analyze the main research themes, the papers were classified into seven categories: Artificial Intelligence and Machine Learning (14 articles), Blockchain (9), Cloud Computing (24), Big Data and Data Analytics (25), ERP Systems (6), Robotic Process Automation (10) and Challenges in Digital Transformation and Construction of SSCs (24). It is important to note that some papers can fit into multiple categories. The final category serves as a residual and catch-all category for miscellaneous studies that do not neatly fit within the other categories. Figure 5 summarizes the results.

Discussion of the findings

The bibliometric and thematic investigation of the literature on Shared Service Center (SSC) digital transformation reveals a rapidly expanding, multidisciplinary field. A content analysis conducted across the full text of the 106 articles paints a striking picture. Research on the digital transformation of Shared Services Centers (SSCs) has grown rapidly since the COVID-19 pandemic, especially in China and Eastern Europe. Despite the growing interest, the topic is theoretically still dispersed and very interdisciplinary, with heterogeneous methodology.

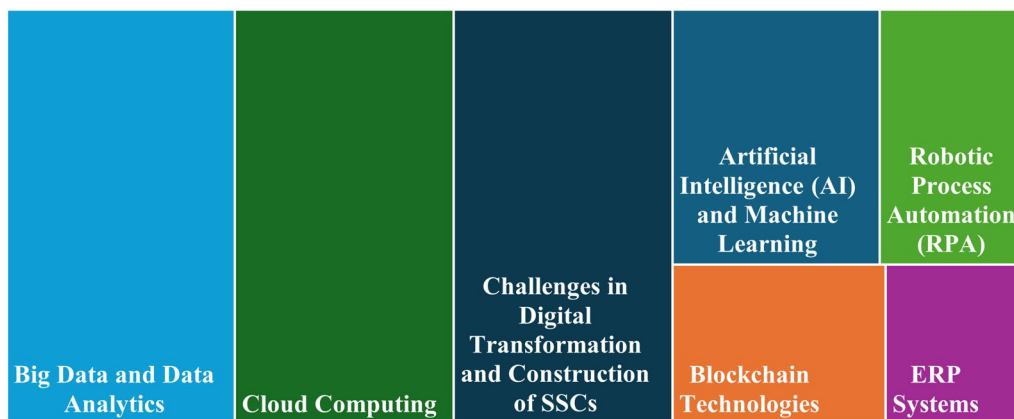


Fig. 5 Main research themes

Emerging technologies in SSCs

The analysis demonstrates that emerging technologies are fundamentally reshaping SSCs from support functions into strategic value centers. Artificial Intelligence and Machine Learning applications show particular promise in enhancing financial management, supporting data-driven decision-making, and increasing operational efficiency while mitigating human error. Using data obtained from a questionnaire answered by 75 heads of finance and accounting in SSCs, Norzelan empirically tested a model based on the Theory of Planned Behavior (TPB) and the Unified Theory of Acceptance and Use of Technology (UTAUT) to explore the adoption of AI [31]. Results showed that performance expectancy attitude, skill, and technical capability are the factors that drive AI adoption in finance and accounting. Grachová concluded that there is no single universal paradigm for the adoption of AI [16]. Thus, the authors distinguish between two models: a) AI integration in business operations with human intervention for key decisions; and b) limited adoption of AI in core operations with reliance on humans for high added value tasks.

The adoption of Artificial Intelligence (AI) within SSCs will also unleash a new era of sophisticated cyber threats that go beyond traditional data breaches, centered on the very basic principles of privacy, ethics, transparency, and accountability [39, 40].

The machine learning models enabling SSC automation manage sensitive HR, payroll, and financial data, but pose significant privacy risks due to their training on large datasets containing personally identifiable information (PII) [16]. Data breaches or model inversion attacks could threaten numerous individuals' private information. The opaque nature of advanced AI algorithms, like deep learning models, raises concerns regarding transparency and accountability in decision-making

processes. Erroneous outcomes from AI, such as incorrectly rejecting payments or tickets, are difficult to trace back to their source, creating a trust deficit and accountability gap regarding who is responsible for errors—data, algorithms, vendors, or internal staff. Moreover, embedded biases in historical training data risk exacerbating discrimination and instigating reputational harm.

Blockchain technologies emerge as another transformative force, enhancing reliability, traceability, and efficiency through its immutable ledger system [6, 24, 53]. Karaszewski et al. [24] presents a case study of an SSC in Elbląg, Poland, that highlights how the combination of cloud solutions and blockchain technology allows for combined electronic document flow and data storage [24]. The shift to a paperless, decentralized document management system reduces resource consumption and waste while lowering storage needs and administrative costs.

Many articles focused on the adoption of cloud technologies in CRM, resource management, and financial data integration. Indeed, cloud-based and “as a service” deployment models have become increasingly prevalent in SSCs. The benefits of cloud platforms include scalability, flexibility, faster time-to-market, agility, and cost efficiency [1, 4, 59]. However, cloud computing also poses significant challenges related to privacy, and as a result, many large businesses and government agencies are reluctant to migrate to a 100% cloud-based deployment model [23].

Big data and analytics are driving performance improvements in SSCs, particularly in financial shared service centers. Recent studies underscore the transformative impact of big data and data analysis in SSCs. Li et al. [27] focuses on performance evaluation within financial sharing centers to improve the performance of financial shared service centers by leveraging advanced

technologies. Meanwhile, [62] focuses on an accounting resource-sharing management system that uses data analysis to streamline operations and improve performance. Wen [58] further contributes by constructing an intelligent financial sharing platform to integrate emerging technologies like 5 G, AI, and big data.

RPA is widely recognized for automating repetitive, low-value tasks, reducing errors, and improving efficiency, especially in the Finance, Accounting, IT, and Human Resources departments [50]. Focusing on the human aspect of the implementation of RPA, [19] ran a questionnaire answered by 175 employees of a Shared Services Center of a large Portuguese construction company and found that employees at all hierarchical levels, including non-management roles, have a positive perception of RPA, with the top management scoring the highest. Additionally, the study found that concerns about the negative impacts of RPA are low. However, [14] explores the implementation of RPA in SSCs through the lens of institution theory and a qualitative approach collecting data via semi-structured interviews from representatives of six major SSCs. The thematic content analysis emphasizes that RPA implementation leads to workforce replacement through practices like early retirements, internal mobility, and reduced outsourced, hinting at future tensions.

Finally, ERP systems play a critical role in standardizing processes and supporting digital transformation in SSCs. A case study of the implementation of ERP in the FSSC at Haier Group, [20, 56] found that this software standardized processes, rules, and languages; optimized its procurement strategy to leverage centralized purchasing to reduce costs. As a result, Haier Group's operating income and net profit increased greatly, and its business scale doubled. Cao et al. [5], Subbarao et al. [51] with questionnaire data from 392 respondents within a Malaysian SSC carried out multiple regression analysis based on the UTAUT model. Results revealed a strong positive link between performance expectancy, effort expectancy, organizational support, and ERP system adoption. However, despite some positive effects, social influence had no statistically significant impact on user adoption.

Overall, the reviewed articles show that the application of emerging technologies in SSCs enables a triple-bottom-line approach—simultaneously achieving operational excellence through secure, automated document flows, financial benefits via reduced administrative costs, and ecological sustainability by minimizing resource consumption. This is remarkably prescient in industries that require strict adherence to data protection standards and audit trails (e.g., finance and insurance) and have a significant ecological impact (e.g., chemical industry).

Challenges in digital transformation and construction of shared services centers

Digital transformation in the context of SSCs is fraught with several multifaceted challenges span technological, human, and organizational dimensions and must be strategically managed to ensure a successful transformation.

Integrating new digital technologies with existing legacy systems is a primary technical hurdle. These outdated systems, often critical to business operations, are frequently incompatible with modern cloud-based platforms and applications [5, 26]. The process requires extensive system reengineering, data migration, and the development of custom APIs or middleware to facilitate communication.

Data fragmentation presents a significant obstacle to effective SSC implementation, particularly in large organizations like State Grid Corporation of China that have struggled with inadequate system integration and siloed data architectures [32]. When data remains trapped in organizational silos without integration, SSCs cannot achieve the process standardization and operational efficiency that justify their existence. This fragmentation often results from organic growth, mergers and acquisitions, or decentralized IT decision-making that has occurred over many years without a unified data strategy. Thus, achieving seamless integration between formerly isolated systems (e.g., ERP, CRM, HRIS) is crucial for automated workflows and requires a master data management strategy to consolidate and cleanse fragmented data.

The digital transformation of an SSC shifts the required skill sets from routine transactional processing to more analytical and technical competencies. This creates a substantial skills gap [7]. SSC managers must navigate dual talent challenges: retention of existing staff through clear career paths and competitive benefits, and acquisition of new talent with expertise in data analytics, automation, and digital technologies [2, 9]. Furthermore, continuous education and training programs must be aligned with industry needs and technological advancements to future-proof the workforce [61].

Digital transformation inevitably induces employee anxiety and stress related to job security, role changes, and new ways of working [14, 19, 20]. Overcoming cultural resistance to centralization and automation is a critical soft challenge. Success depends on effective change management strategies that include transparent communication, strong leadership endorsement, and involving stakeholders in the design process. Managing the emotional impact and fostering a culture open to innovation and continuous adaptation is essential for smooth implementation.

Finally, navigating all these challenges requires a clear strategic vision and bold leadership. A common challenge is defining and tracking the right key performance indicators (KPIs) that go beyond cost reduction to measure value creation, quality, efficiency, and customer satisfaction. Without this strategic alignment, the SSC risks being viewed as a mere cost center rather than a strategic partner contributing to operational excellence, financial benefits, and ecological sustainability.

Controversies and gaps in the literature

There are several controversies within the reviewed literature. While most authors agree that the process of digital transformation is inevitable, the level of optimism varies significantly across a variety of issues: the pace of technological change; security and privacy; and the impact of emerging technologies in the workplace.

Some authors want to maintain or even accelerate the pace of digital transformation arguing that this process is inevitable and it is crucial to maintain a competitive edge [58]. However, other authors have expressed concern that the accelerated pace of technological change can lead to instability and disruption within organizations, a competency crisis, and anxieties about job displacements [33, 35, 36, 47, 52].

On the topic of security and privacy, tech optimists emphasize how advanced technologies like blockchain can guarantee data immutability, and enhance cyber security and data encryption [15]. In addition, these authors argue that emerging technologies like AI and Cloud Computing enable and facilitate corporate compliance with an increasingly complex body of regulations [56]. In contrast, some authors argue that the concentration of sensitive information in SSCs increases the cybersecurity risk [46].

Tech optimists emphasize the potential of automation and AI to vastly improve the levels of labor productivity [19, 21]. By automating repetitive and monotonous tasks, human employees can focus on higher value-added tasks that are more engaging. These authors show evidence that most employees and managers in SSCs have a favorable view of automation and argue that the risks of job displacement are exaggerated. Nonetheless, several academics are deeply concerned about the risk of AI causing workplace disruption and job losses [14, 60], arguing for measures to mitigate these risks.

Finally, several gaps within the literature were identified. Most studies did not explicitly mention any of the United Nations Sustainable Development Goals or sustainability initiatives. For instance, although technologies like cloud computing and blockchain may lead to greater sustainability, these connections remain largely implicit and underexplored in the literature. In addition, the

majority of studies analyzed focused on the Asia-Pacific region, particularly the PRC. As a result, there is a notable scarcity of research conducted in regions like Southern Europe, Latin America and the Caribbean, Africa, and the Middle East.

Main conclusions

To sum up, this scoping review of the literature available in 5 databases (IEEE Xplore, Web of Science, SCOPUS, OpenAlex, and JSTOR) and published over the last decade sought to map out the landscape of digital transformation of shared services centers. Emerging technologies are transforming Shared Services Centers (SSCs) from support functions into strategic, value-driven hubs. AI, cloud computing, RPA, and blockchain enhance operational efficiency and scalability but introduce significant challenges in cybersecurity, ethical governance, and workforce adaptation. These technologies enable more automation, improved data integration, and more informed decision-making processes while transforming operational efficiency and scalability.

Implementing these technological solutions also presents extremely important challenges that must be carefully managed. These involve cybersecurity threats with centralized sensitive data, ethical governance related to automated decision-making systems, and significant workforce adjustment requirements. Further challenges include the complex integration of new technologies with legacy systems, talent management issues in competitive digital economies, and employee resistance to technological change. It identifies substantial research gaps in aligning digital SSC strategies to overall sustainability objectives and global development agendas.

Practical implications of the study

The study offers practical value to researchers, professionals, and decision-makers by systemically mapping the digital transformation environment of Shared Services Centers (SSCs). For researchers, it marks clear thematic and geographical lacunae—e.g., the absence of reference to ethical AI, cybersecurity, and sustainability-related SSC strategies—allowing future research to be more concentrated and productive. By bringing together dispersed knowledge, the review helps scholars position their research in the broader scholarly conversation and defend novel contributions.

For corporate leaders and SSC managers, the findings provide an evidence-based framework for strategic decision-making. The technology groupings (e.g., blockchain, AI, RPA) and associated concerns offer a pragmatic roadmap for prioritizing digital initiatives, managing risks by implementation, and aligning SSC capabilities with sustainability imperatives like SDGs 8, 9, and 12. Moreover,

encouraging best practices and regional success—particularly from high-performing SSC regions—can feed into operational planning and policy creation.

Finally, the review calls for integrating theoretical models and actual digital transformation initiatives to enable organizations to develop stronger, ethical, and sustainable SSC models. These implications offer a roadmap for leveraging SSCs as innovation drivers, sustainability forces, and strategic levers in a digitally transformed world.

Limitations and scope for future research

Finally, it is worth considering the limitations of this scoping review. Firstly, this review was confined to articles from 5 databases whose full texts were accessible. As a result, many papers from closed-access journals and conferences potentially result in the omission of relevant articles. Secondly, because the search was carried out in July of 2024 and only included articles in the final stage of publication, it was impossible to take into consideration all the publications for the year 2024. Given that the majority of research was done in China and Eastern Europe, future research should focus on the experiences in regions like Iberoamerica. Likewise, scholars should seek to align their research agenda with the SDGs and explore innovative use cases for emerging technologies within the context of SSCs.

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Author contributions

A.S.P.: Supervision, Writing - Reviewing and Editing. A.A.A.: Supervision, Writing - Reviewing and Editing. J.P.: Conceptualization, Methodology, Data curation, Writing. M.P.C.: Supervision, Writing - Reviewing and Editing. GPT-4 (ChatGPT): Language enhancement, Formatting, Plagiarism check.

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Data availability

No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest

The authors declare that they have no Conflict of interest.

Ethics approval

Ethical approval for this study was granted by the CEOS:IPP Ethics Committee.

Consent for human participants

Informed consent was obtained from all participants involved in this study. All participants were fully informed about the study's aims, procedures, and potential risks before agreeing to participate.

Consent to publication

All authors have consented to the publication of this paper and its contents. Furthermore, all participants have provided their consent for their data to be published.

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